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Research Statement

My research combines the data-rich nature of eye-tracking analytics with collaborative work in augmented reality to develop novel methods for co-located communication. Co-located collaboration is prevalent across many domains, particularly in contexts that require interdisciplinary work and asymmetric expertise. Augmented reality headsets provide a unique opportunity to support communication and the understanding of complex concepts through 3D visualizations. However, current head-mounted displays can be cumbersome and may obstruct natural social interactions, such as eye contact and facial expressions. Eye contact is a fundamental social phenomenon that helps collaborators understand each other's attention and intentions.

When these crucial social cues are disrupted in augmented reality, social presence is diminished, reducing the quality of interaction and cooperation. Shared-gaze visualizations (SGV) help overcome these limitations by enabling collaborators to rely on natural gaze interactions while also enhancing communication. By tracking collaborators' eye-gaze directions in real time, we can provide visualizations that reveal their partner's areas of interest and focus (see laser visualization in Figure 1).

My primary academic interests are advancing methodologies for shared-gaze visualizations in augmented reality. Specifically, I investigate methods for: 1) broadening approaches to visualization design, 2) identifying the functional limitations of visualizations in specific tasks, and 3) understanding how users' perceptions shift in response to modality limitations.

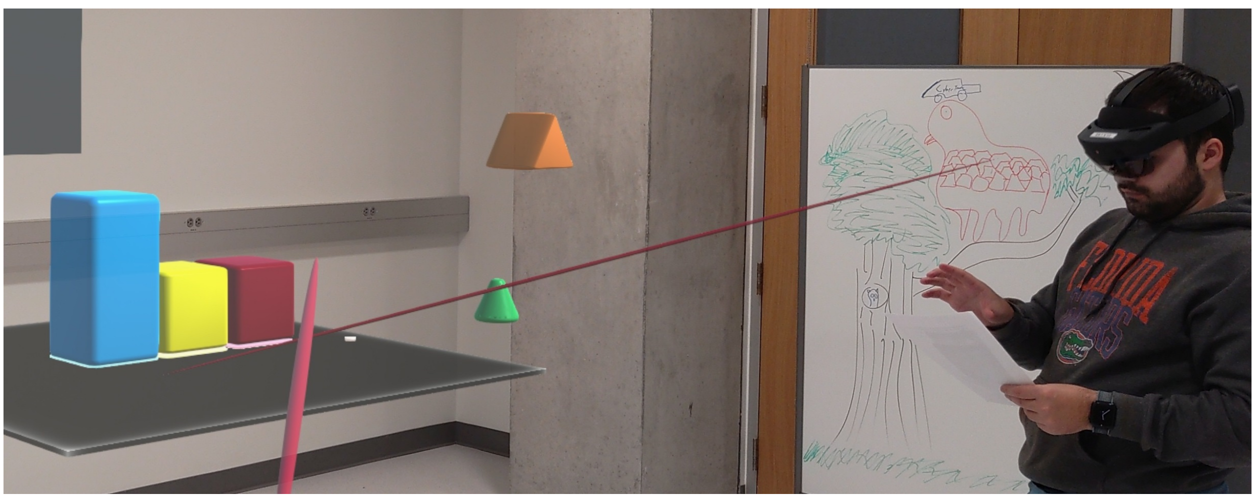

Figure 1: POV of shared-gaze visualization laser ray in a simulated assembly task in augmented reality.

Visualization Design

Shared-gaze visualizations can create meaningful social experiences in augmented reality headsets by allowing collaborators to engage with each other while simultaneously interacting with virtual artifacts. Current implementations, such as laser rays (Figure 1), present several limitations. They raise privacy concerns, as users are always aware of their partner's focus; prior evaluations have shown no clear distinction between user preferences and performance; dynamic visualizations are often difficult for users to interpret; and most designs lack mechanisms for user-controlled activation.

To address these limitations, I developed two new approaches for visualizing gaze. Building on the existing laser ray implementation, I created a method that uses a user's gaze dwell (1.7s) to trigger activation of the ray. If the user shifted their focus, the visualization disappeared.

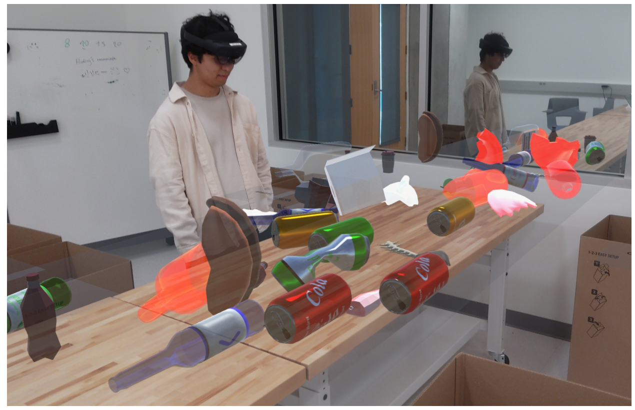

Figure 2: Dynamic sorting task simulation using a physical table base and cardboard boxes..

The second approach was a user-detached method for visualizing direct eye-gaze: highlighting objects of interest (see Figure 4). In this method, a user's gaze direction and orientation determined which objects they were observing, activating a bright change in the object's color hue [6]. This approach allowed collaborators to perceive their partner's focus while preventing constant monitoring of their own gaze, thereby preserving privacy and situating visualizations within the task context.

An evaluation of the two methods showed strong potential for advancing SGV design methodologies, but also underlined limitations, particularly with the highlight visualization: users reported difficulty focusing on objects due to hue changes. This work was published and presented at *IEEE VR 2024* as a poster paper and received the *Best Poster – Honorable Mention* award.

Building on the previous gained insights, I investigated how restricting visualizations to one partner versus both (see Figure 4) could reduce visual clutter in dynamic sorting tasks [7]. While this approach did not help mitigate distraction, it revealed participants felt they were excluded from the task when their gaze was not visualized by their partner's was. My findings highlighted the importance of social presence within a virtual task. This work was published at *IEEE ISMAR 2025* as a conference paper and supported in part by the National Science Foundation.

Figure 3: Gaze outline laid over a glass bottle simulated in augmented reality.

Drawing on these findings, I developed a novel method that uses collaborators' gaze direction to outline the borders of an object (Figure 3). This approach allowed collaborators to perceive object characteristics clearly while still maintaining a sense of privacy [5].

Contextual Design

How information is presented in an interface varies greatly depending on the context in which the interface is used. We can best support collaborators in a co-located space with shared-gaze visualizations by understanding what types of visualizations are most appropriate for specific contexts.

My work has examined how SGVs are interpreted in scenarios that reflect the very contexts they are designed to support: assembly, sorting, and manufacturing. In contrast, prior evaluations of SGVs have largely been conducted in abstract settings that fail to simulate applied contexts.

These studies often lacked external stimuli, offered limited interaction modes, and relied on static implementations. Recognizing these gaps in visualization evaluation, I led research exploring how interacting with instruction manuals influences user perceptions of SGVs during assembly tasks (Figure 1). The findings revealed that dynamic visualizations can become distracting when users' attention is divided across multiple demands [6].

Following our study of SGVs in a static assembly context, I led a research team to evaluate SGV designs in a dynamic sorting task using a simulated conveyor belt (Figure 2). In this study, we also incorporated audio stimuli alongside the virtual artifacts. Most prior evaluations of SGVs have been conducted in controlled environments without external factors such as audio noise. Studying SGVs under conditions where collaborators are constrained to rely heavily on visual cues provides a deeper understanding of how these tools influence interaction and allows their effects to be more clearly isolated.

Accordingly, we evaluated the outline visualization in a setting that simulated the noisy environment of a recycling plant. The recycling plant scenario was chosen because it involves sorting potentially hazardous objects and requires close coordination among collaborators in a high-stress environment. Results from the evaluation showed that the outline visualization both improved participants' ability to recognize their partner's intentions and was rated as significantly more engaging compared to other visualizations when audio noise was present [5].

This work was published as part of my doctoral dissertation at the University of Florida and was supported in part by the McKnight Doctoral Fellowship and the National Science Foundation.

Research Directions

What can gaze interactions tell us about social dynamics? A wealth of information is conveyed through non-verbal communication, particularly through gaze cues and eye contact [3]. A person's gaze direction can signal to others in a social setting whether they are paying attention, interested in a topic, or even dissatisfied. By analyzing eye-gaze interactions, such as direction, position, dwell, focus, and regions of interest, in group contexts, we can develop methodologies for interpreting ongoing social dynamics. I am particularly interested in how different tasks shape group eye-gaze interactions, especially within high-stakes contexts like industrial and medical applications. In these high-stress environments, verbal communication is often constrained or unavailable due to environmental limitations. Such constraints create a need for alternative modes of communication that remain reliable under pressure. My research aims to uncover patterns of gaze behavior associated with well-functioning teams, as well as those prevalent in teams that lack cohesion.

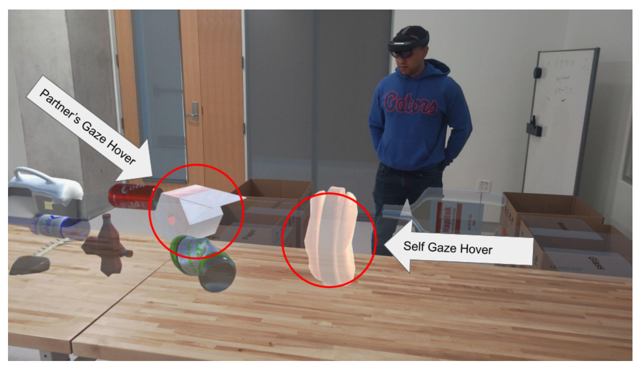

Figure 4: Hover visualization with partner.

Adaptive frameworks. While understanding how eye-gaze interactions shape our view of group dynamics, I am also interested in exploring how this knowledge can inform adaptive frameworks within existing systems. In particular, I focus on shared-gaze visualizations and the use of visual cues to support communication among collaborators in augmented reality settings. My goal is to identify which types of visualizations align best with specific levels of stress, workload, and group dynamics.

Medical applications. My research has primarily focused on generalized applications such as assembly and sorting. My future work will extend to understanding the use of shared-gaze visualizations in medical contexts. Prior studies have shown that gaze sharing and virtual pointers can reduce workload and increase task success in surgical training scenarios by clarifying ambiguous verbal cues. However, SGVs still lack clinical applicability: visualizations can be distracting [4], often lack sufficient granularity to determine thresholds where shared gaze ceases to be beneficial, or even becomes detrimental, for specific tasks, and rarely employ adaptive methods to tailor visualization strategies to context [2]. I plan to explore surgical applications through contextual inquiries, using simulated scenarios to capture user perspectives and actions, with the goal of developing specific guidelines tailored to the needs of focused user groups [1].

To conclude, my ultimate goal is to bridge the gap between natural eye-gaze interactions, the understanding of group dynamics, and the development of adaptive frameworks for tailored human–computer interaction in collaborative augmented reality. At the same time, I plan to explore how these tools can be most effectively applied to support professionals in industrial and medical contexts. I aim to pursue funding through national agencies including NIH, NSF, and DARPA.

During my graduate studies, I had the fortunate opportunity to collaborate with more than 30 authors and researchers across the country, and I intend to continue this trend throughout my professional career.

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